



# Welfare

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# Outlines

- Welfare
- Pareto Efficiency
- Aggregation of Preferences
- Social Welfare Functions
- Welfare Maximization
- Individualistic Social Welfare Functions
- Fair Allocations
- Envy and Equity



# Welfare

Social welfare refers to government programs and initiatives designed to promote the well-being of society as a whole, particularly focusing on those who may be disadvantaged, marginalized, or in need of assistance. These programs aim to enhance the quality of life for individuals and communities by providing various forms of support, such as financial aid, healthcare services, education assistance, housing assistance, food assistance, and employment training

# Government Social Welfare Programs



- Health
- Education Assistance
- Financial Assistance
- Unemployment Insurance
- Housing Assistance

# Pareto Efficiency

- Pareto efficiency, also known as Pareto optimality, is a concept in welfare economics that describes a situation in which it's impossible to make one individual better off without making at least one other individual worse off. In other words, an allocation of resources is Pareto efficient if there is no way to reallocate those resources that would improve the welfare of one person without reducing the welfare of someone else

# Aggregation of Preferences

- Aggregation of preferences is the process of combining individual choices or rankings into a collective decision, often involving complex considerations of fairness, efficiency, and the resolution of potential conflicts or paradoxes.

# Voting Method of Aggregation of Preferences

- The voting method of aggregation of preferences involves individuals expressing their preferences through votes, with the collective decision determined by the outcome of the voting process, which can vary depending on the specific voting system employed, such as plurality voting, ranked-choice voting, or proportional representation.

# Problem

- This method may not generate transitive social ordering.





# Rank vs Order Voting

- Rank voting involves individuals assigning specific ranks to options in order of preference, while order voting simply requires individuals to list options in a preferred order without assigning specific ranks.

# Limitation of Rank Voting

- One potential problem associated with the voting method of aggregation of preferences is the possibility of strategic voting or manipulation, where individuals may vote strategically rather than honestly to influence the outcome in their favor, leading to results that do not accurately reflect their true preferences.

# Arrow's Impossibility Theorem

- Arrow's Impossibility Theorem states that there is no perfect voting system that can avoid all possible issues when aggregating individual preferences into a collective decision.
- An example of Arrow's Impossibility Theorem could be illustrated in a simple voting scenario where a group of individuals with different preferences must decide on three options: A, B, and C. If the majority prefers A over B and B over C, but a minority prefers C over A, no voting system can accurately represent all preferences while still satisfying Arrow's conditions.
- For 3 properties of Arrows theorem see page # 655 of text book.

# Social Welfare Function

- A social welfare function is a mathematical representation used in welfare economics to aggregate individual welfare or utility into a collective measure of societal well-being.

# Social Welfare Function

- A basic form of a social welfare function  $W$  can be represented as follows:

$$W = f(u_1, u_2, \dots, u_n)$$



- Where:
- $W$  represents the collective societal welfare or utility.
- $f$  is the social welfare function, which is typically a mathematical function or rule.
- $U_1, U_2, \dots, U_n$  are the individual welfare or utility levels of  $n$  individuals in the society.

# Classical Utilitarian or Benthamite

Classical Utilitarian  $W = \sum u_i$  The societal welfare is the sum of the individual welfare or utility

# Weighted Sum of Utilities Welfare Function

- $W(U_1, \dots, U_n) = W = \sum a_i u_i$
- *Where  $a_1, \dots, a_n$  weights assign*



# Minimax or Rawlsian SWF

- $W(U_1, \dots, U_n) = \min(U_1, \dots, U_n)$
- This welfare function says the social welfare of an allocation depends only on the welfare of worst off agent.

# Welfare maximization

- Welfare maximization refers to the goal of maximizing the overall well-being or utility of individuals in a society, typically achieved through policies, programs, and decisions that aim to improve living standards, reduce poverty and inequality, promote social justice, and enhance overall quality of life.
- An Allocation that max welfare must Pareto efficient
- See diagram on page 658 and 659

# Welfare Max

- $\max W(u_1(x), \dots, u_n(x))$
- $\sum X_i = X_1$
- *Read Page 657 of Text Book*

# Individualistic Social Welfare Functions,

- $W=W(U_1(x_1), \dots, U_n(X_n))$
- Also known as the Bergson-Samuelson Welfare Function

# Fare Allocation

- "Fair allocation" typically refers to the equitable distribution of resources, opportunities, or benefits among individuals or groups in a society, often guided by principles of justice, equality, or need.
- For detail discussion see text book page #660

# Envy and Equity

- Symmetric or Equitable
- The allocation is equitable if no agent prefer any other agent's bundle of goods to his or her own.
- Envy
- If some agent  $i$  does prefer some other agent  $j$ 's bundle of goods, we say that  $i$  envies  $j$ .
- Fair
- Finally if an allocation is both equitable and Pareto efficient, we will say that it is a fare.



Thank

You